Statistical Analysis of Physical Systems using Monte Carlo and MCMC

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OBJECTIVE

The objective of this project was to understand and simulate physical systems using **Markov Chain Monte Carlo (MCMC)** methods. Specifically, we implemented the **Ising Model** to observe how microscopic interactions between spin states lead to macroscopic phase transitions in magnetic materials.

The first assignment was assigned to make us familiar with coding and how to translate the- oretical concepts of thermodynamics into code to solve problems. The primary language used for these assignments is Python as it's easy to get started with and relatively simple to implement. Since I was already familiar and comfortable with coding in python, the next task was to brush up my understanding of thermodynamic concepts.

In the second assignment, we covered the basics of Probability and Statistics where we briefly covered topics of distributions and comparing them to derive insights from the data as well as the law of large numbers.

Contents

1 Theoretical Background

1.1 Markov Chains and MCMC

A Markov Chain is a stochastic model where the probability of a future state depends solely on the current state (the Markov property). MCMC allows us to sample from complex probability distributions, such as the Boltzmann distribution, by generating a chain of states that converges to equilibrium.

1.2 The Metropolis-Hastings Algorithm

This algorithm is the engine of our simulation. We propose a random spin flip; the move is accepted if it reduces energy ($\Delta E < 0$), or with probability $e^{-\Delta E/k_B T}$ if it increases energy. This ensures the system explores states proportional to their physical probability.

1.3 The Ising Model and Phase Transitions

The Ising model represents a lattice of spins $s_i \in \{+1, -1\}$. The Hamiltonian is:

$$H = -J \sum_{\langle i,j \rangle} s_i s_j$$

At the critical temperature (T_c), the system undergoes a spontaneous symmetry breaking—a phase transition from a disordered paramagnetic state to an ordered ferromagnetic state.

2 Implementation and Observations

Our implementation used the Metropolis algorithm to evolve the lattice.

- **High Temperature ($T > T_c$):** Thermal fluctuations dominate; spins flip randomly, leading to a zero average magnetization.
- **Low Temperature ($T < T_c$):** Spin-spin interaction (J) dominates, leading to the formation of large, ordered spin domains.

3 Conclusion

This project demonstrated that complex macroscopic behaviors, such as phase transitions, emerge directly from simple local rules. We successfully verified the efficacy of MCMC as a computational tool for statistical mechanics, observing the transition from disorder to order as the system cools below its critical temperature.

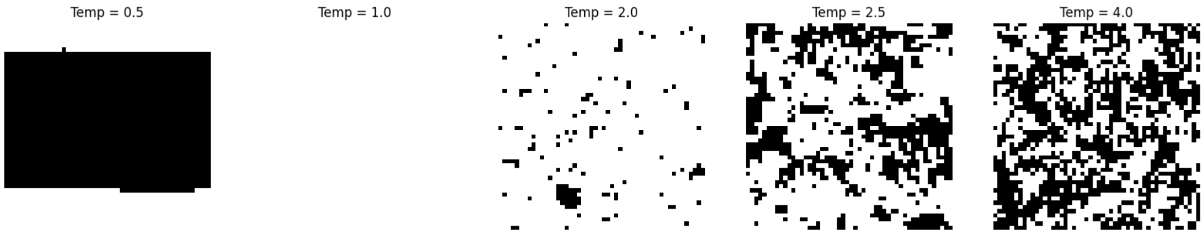

Figure 1: Spin patterns at different temperatures

Figure 2: Spin Organization Over Time (T = 0.5)

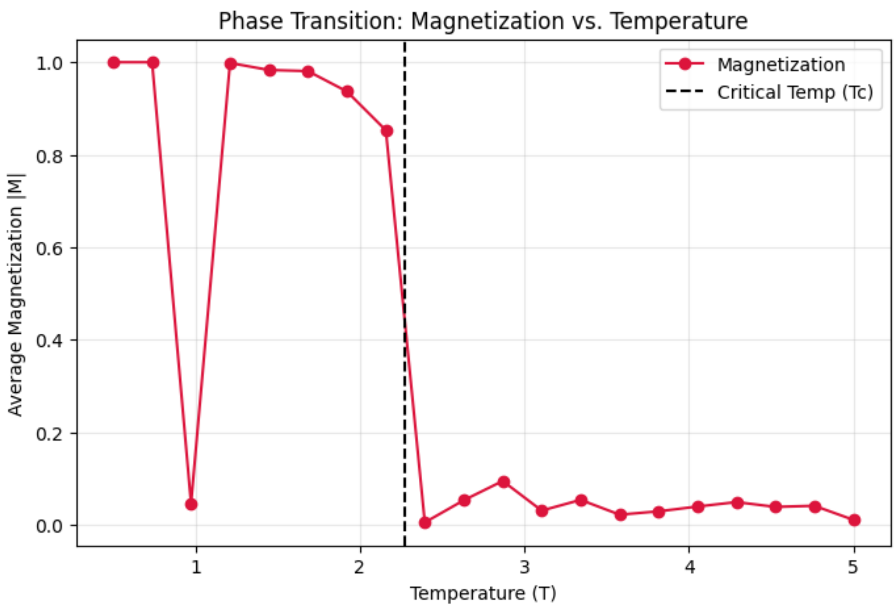

Figure 3: Phase Transition: Magnetization vs. Temperature